INTERNATIONAL JOURNAL OF HYDROGEN ENERGY 44 (2019) 5573–5584

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Experimental investigation of combustion characteristics and NOx emission of a turbocharged hydrogen internal combustion engine

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ARTICLE INFO

Article history:
Received 29 July 2018
Received in revised form
14 August 2018
Accepted 28 August 2018
Available online 9 October 2018

Keywords:
Turbocharged
Hydrogen engine
Combustion characteristic
NOx emission

ABSTRACT

In order to alleviate the contradictions of increasingly prominent environmental pollution, greenhouse gas emissions and oil resource security issues, the search for renewable and clean alternative energy sources is getting more and more attention. Hydrogen energy is known as a future energy source because of its safety, reliability, wide range of resources and non-polluting products. Hydrogen internal combustion engine combines the technical advantages of traditional internal combustion engines and has comprehensive comparative advantages in terms of manufacturing cost, fuel adaptability and reliability. It is one of the practical ways to realize hydrogen energy utilization. In this paper, the combustion characteristics and NOx emission of a turbocharged hydrogen engine were investigated using the test data. The results showed the combustion duration (the crank angle of 10%–90% fuel burned) at 1500 rpm and 2000 rpm was equal and the combustion duration is much bigger than the other loads when the BMEP is 0.27 MPa. The reason is the effect of the turbocharger on the gas exchange process, which will influence the combustion process. The cylinder pressure and pressure rise rate were also investigated and the peak pressure rise rate was lower than 0.25 MPa/°CA at all working conditions. Moreover, the NOx emission changed from 300 ppm to 1200 ppm with engine speed increasing and the maximum value can reach to 7000 ppm when the equivalence ratio is 0.88 at 2500 rpm, maximum brake torque. The NOx emission shows different changing tendencies with different working conditions. Finally, these conclusions can be used to develop controlling strategies to solve the contradictions among power, brake thermal efficiency and NOx emission for the turbocharged hydrogen internal combustion engines.

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Introduction

According to the target of COP21 meeting to keep global warming below 2 °C in Pairs in 2015, the carbon dioxide (CO_2) emissions should be cut by 60% by 2050 of the world [1]. The way of decreasing the carbon can be summarized as two aspects: improving the thermal efficiency of traditional energy and looking for alternative energy sources. Another problem

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https://doi.org/10.1016/j.ijhydene.2018.08.184
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Nomenclature

BTE	brake thermal efficiency
EGR	exhaust gas recirculation
λ	equivalence ratio
T	temperature, K
TDC	top dead center
WOT	Wide open throttle
ST	spark timing
n	engine speed, rpm
HICE	hydrogen internal combustion engine
MBT	maximum brake torque

is the shortage of traditional energy, which will also limit the economic development in the not far future. Thus, looking for the no carbon or less carbon, sustainable and renewable energy is getting more and more attention. The Hydrogen Council, which is an alliance of 18 companies, showed the effect of hydrogen on the decreasing the CO_2 in the COP23 meeting in Bohn [2]. Moreover, hydrogen is a colorless, odorless and zero (harmful) emission fuel when burned with oxygen. Hydrogen can be got in many ways, such as natural gas reforming, gasification using coal or biomass, electrolyzing water, high-temperature thermochemical water-splitting, renewable liquid reforming, direct solar water splitting, microbial hydrogen production and biological processes and so on. Thus, the development of hydrogen is one of the best ways to deal with global warming and shortage of conventional energy [3].

Utilizing the hydrogen mainly has two ways: hydrogen fuel cells and hydrogen internal combustion engines. Hydrogen fuel cells can achieve high efficiency (around 60%) and zero emission, which are the two advantages [4]. Moreover, the noise of the reaction between hydrogen and oxygen is very small, which helps improving the comfort of the vehicles. However, hydrogen fuel cells still exists several problems mainly including: the low efficiency of the total system (around 48%–52%), the high price, the low reliability and the high requirements of fuel purity (over 99.999%). These problems limit the promotion and application of fuel cells [5]. However, hydrogen internal combustion engines can utilize the advantages of the traditional engines, which has been developed in the two hundred years. The hydrogen engine can not only get the high brake thermal efficiency (BTE) because the fast combustion velocity of the hydrogen-air mixture, but also solve the problems of the price, reliability and variable working conditions. Furthermore, the hydrogen engine has no demand to the hydrogen purity. Thus, hydrogen internal combustion engine is an easy way to realize the using of hydrogen.

The biggest problem of hydrogen engine is the contradiction among power density, BTE and NOx emission. Increasing the intake air pressure can increase the power density and BTE because it can rise the intake air flow mass and improve the gas exchange process. A 6.8 L V-10 and a 2.3 L I-4 hydrogen engine were investigated by Ford Groups [6,7]. The results showed the torque of the supercharged hydrogen engine can reach to the level of the compressed natural gas engines. BMW Groups also reported a 0.5 L supercharged hydrogen engine with 0.185 MPa of the intake air pressure [8] and they found that the mean effective pressure can overtook 1.8 MPa for all working conditions with variable compression ratios. Verhelst et al. [9] developed the GM 7.4 L V8 supercharged hydrogen engine and the maximum torque was 60% higher than the naturally aspirated hydrogen engines. Moreover, they explored the way of stoichiometric mixtures add with exhaust gas recirculation (EGR) can be got a higher BTE than the lean equivalence ratio and the power can increase 30% than the basic gasoline engine [10]. Natkin reported the operation of a supercharged 2.3 L hydrogen engine at equivalence ratio (λ) = 0.5 [11]. The results showed the maximum torque of the hydrogen engine was smaller by 28% than the basic gasoline engine. Ford Groups introduced a 4.0 L hydrogen engine with a turbocharger and the maximum can reach to 35.5% [12]. Das et al. [13] presented a 2.5 L turbocharged hydrogen engine with an intercooler, which can obtain 66.7 kW at 3200 rpm and the NOx emission was over 800 ppm though the equivalence ratio was controlled smaller than 0.6. Thus, boosting the intake air pressure can increase power density and BTE while the NOx emission will increase.

Studying the combustion characteristics of hydrogen engines is an effective approach to increase the comprehensive performance and avoid the abnormal combustion phenomenon. Yamin et al. [14] utilized the simulation calculations for comparing and analyzing the combustion duration of hydrogen and gasoline engines. They found that the combustion duration of hydrogen engine was shorter than that of gasoline engines and it decreased with the increase of equivalence ratio. Using the zero-dimensional model, Baghdadi et al. [15] explored the effects of compression ratio, equivalence ratio and engine speeds on the power, BTE, NOx emission characteristics and pre-ignition limit of the hydrogen engine. Ma Jie et al. [16] used a zero model to calculate the performance of a hydrogen internal combustion engine and optimize the structural parameters of the engine. They results showed that, the combustion pressure increased and the thermal efficiency decreased as the equivalence ratio increasing. Ma Fanhua et al. [17] explored the variation of combustion parameters of hydrogen internal combustion engines under different working conditions on a single-point injection hydrogen engine. The research showed that the hydrogen internal combustion engine can complete combustion at different speeds, while the impact is not as great as the equivalence ratio. Junfa Duan et al. [18] explored the NOx emissions mechanism of the hydrogen engine under high load using CONVERGE software. The Simulation results showed that the equivalence ratio can increase the temperature inside the flame front and the OH concentration in the flame front. There are a number of researchers study on the combustion characteristics of the mixture of the other fuel and hydrogen [19–23]. These results were investigated in the naturally aspirated hydrogen engines and didn't take about the effect of boost pressure on the combustion characteristics. Das et al. [13] introduced the several combustion characteristics of turbocharged hydrogen engines such as combustion duration, cylinder pressure and peak cylinder pressure with the engine speeds. However, in order to improve the performance of turbocharged hydrogen engine, the effect of engine speeds,

loads, equivalence ratios and ST on the power and BTE is not only needed to be studied, but their effect on the combustion characteristics and NOx emission also is needed. The author has reported the effects of the different parameters on the power and BTE in Reference [24,25] and this paper explored the combustion characteristics and NOx emission with variations of engine speeds, loads, equivalence ratios and spark timing. These results can be used to propose the comprehensive strategies to solve the contradictions among the power, BTE and NOx emission for hydrogen engines.

Material and method

To explore the combustion characteristic and NOx emission of the turbocharged hydrogen engines, a 2.3 L, 4-St, port fuel injection gasoline engine was converted to a hydrogen engine. The basic parameters and the uncertainty and range of the sensors are listed in Tables 1 and 2. The sketch of the hydrogen engine is shown in Fig. 1.

The test is carried out using the FC2000 test system in this paper, which includes a CW250 eddy current dynamometer, oil and water cooling system, intercooler system for the turbocharger and data record system. The eddy current dynamometer is used to absorb power and motor. Cooling system is used to control the temperature of the oil and cooling water through exchanging the heat with the water. Intercooler for the turbocharger is used to control the temperature of the air compressed by the compressor, which also exchanges the heat with the water. A hot line air mass flow meter is used to measure the air mass flow, which is listed in Table 2. Two kinds of hydrogen mass flow meters are used to measure the hydrogen mass flow, whose types are CMF010 and CMF025 of Emerson Groups. AVL DiGas 4000 Light is used to measure the concentration of NOx, CO, CO_2 and HC emission in the exhaust gas. The cylinder pressure is measured by the transient pressure sensor of Kistler 6118 type and the data is recorded by the combustion analyzer of Kibox. Moreover, the crank angle is measured by the encoder of Kistler 2614C type. The data is recorded by the computer and used to analyze the combustion characteristic of the turbocharged hydrogen engine.

To ensure the accuracy and repeatability of the test data, the hydrogen engine should be warmed up to the operating temperature. General speaking, the oil temperature of 90 °C and the cooling water temperature of 80 °C are the standard temperature to conduct the test. Working conditions to study the effect on the combustion characteristic and NOx emission can be divided into four parts as shown in Table 3. The engine speeds from 1500 rpm to 4000 rpm with interval of 500 rpm and parts of loads were presented for the study. Moreover, the different equivalence ratios and spark timings (STs) also were showed.

Table 1 – Parameters of the 2.3 L turbocharged hydrogen engine.

Engine type	Hydrogen engine
Bore	90 mm
Stroke	90 mm
Cylinders	4
Crank radius	45 mm
Connecting rod radius	153 mm
Compression ratio	9.3
Exhaust valve open	53.6°BBDC
Exhaust valve close	44.4°ATDC
Inlet valve open	23.3°BTDC
Inlet valve close	58.7°ABDC
Maximum torque @ 3000 rpm	161 N.m
Maximum power @ 4000 rpm	59.4 kW

Table 2 – Average uncertainty of the main measured parameters.

Parameter	Uncertainty	Range
Engine speed	±1	0–8000 rpm
Torque	±0.4%FS[a]	0–1100 N.m
Pt100	±0.5	0–423 K
Thermocouple K type	±1	0–1473 K
Intake pressure	±0.4%FS[a]	−0.15-0.25 MPa
Exhaust pressure	±0.4%FS[a]	0–0.1 MPa
Oil mass flow rate	±0.5%FS[a]	0.6–6 m^3/h
Cooling mass flow rate	±0.5%FS[a]	1.5–15 m^3/h
Air mass flow rate	±1% FS[a]	0–450 kg/h
Hydrogen mass flow rate	±0.1% FS[a]	0–6.5 kg/h

FS[a]: Full scale

Result and discussion

Effect of engine speed variations

The engine speed has an effect on the gas exchange process, the reaction time of air and fuel in the cylinder and the hydrogen injection process. Especially with a turbocharger, the turbulence intensity in the cylinder increases with the increase of the engine speed, which will make a significant differences in the combustion process and NOx emission in the cylinder compared with the naturally aspirated engine. As Fig. 2 shown, the pressure at the end of compression stoke increases with engine speed increasing, which can change from 1.7 MPa at 1500 rpm to 2.6 MPa at 4000 rpm but the increments decrease when the engine speed is bigger than 2500 rpm. The reason can be explained from two aspects: on the one hand, the intercooled and boost pressure increases with engine speed increasing as Fig. 3 shown, resulting in the increase of the pressure at the end of compression stoke. On the other hand, the valve phase and the intake manifold are designed to achieve the maximum volumetric efficiency at 3000rpm-3500 rpm and the air flow into the cylinder is increased, which lead to the increase of the pressure at the end of compression stoke. The increase of pressure at the end of compression stoke will lead to the increase of the peak cylinder pressure as Fig. 4 shown. The peak cylinder pressure decreases at 4000 rpm because the boost pressure is controlled smaller than 0.14 MPa to avoid knock [28,29] and the volumetric efficiency also decreases when the engine speed is bigger than 3500 rpm. Thus, the peak cylinder pressure decreases at 4000 rpm.

The peak cylinder pressure rise rate is closely related to the combustion noise and is an important parameter to

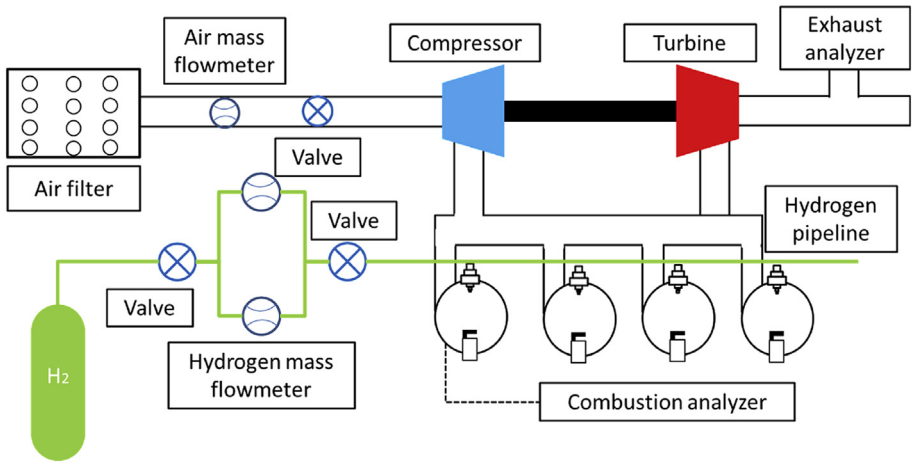

Fig. 1 – Sketch of the turbocharged hydrogen internal combustion engine.

Table 3 – Measurement points.

Title	Range	Step
(1) Engine speed and load variations		
Engine Speed	1500 rpm-4000 rpm	500 rpm
Load	Minimum- wide opening throttle (WOT)	2 deg
λ	0.55	–
ST	Maximum brake torque (MBT)	–
(2) Equivalence ratio variations		
Engine Speed	2500 rpm	–
Load	25%	–
λ	0.4–1.1	0.1
ST	MBT	–
(3) Spark Timing variations		
Engine Speed	2500 rpm	–
Load	50%, 75%	–
λ	0.55	–
ST	-5-25 °CA BTDC	5°CA

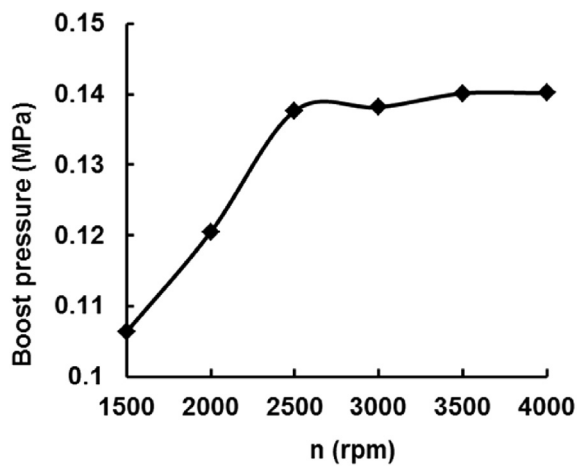

Fig. 3 – The intercooled and boost pressure changes with engine speeds at WOT.

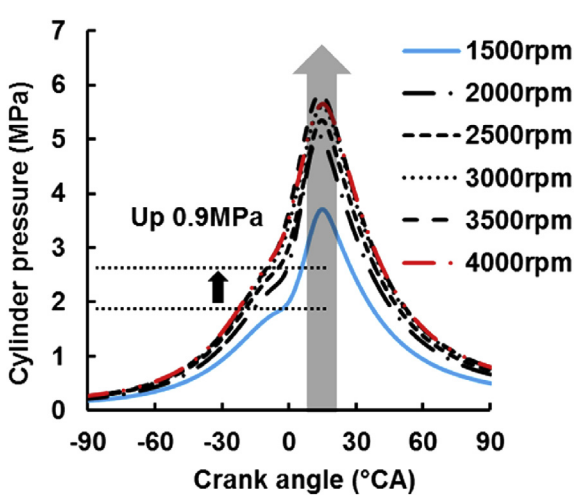

Fig. 2 – The cylinder pressure changes with crank angle at different engine speeds, WOT.

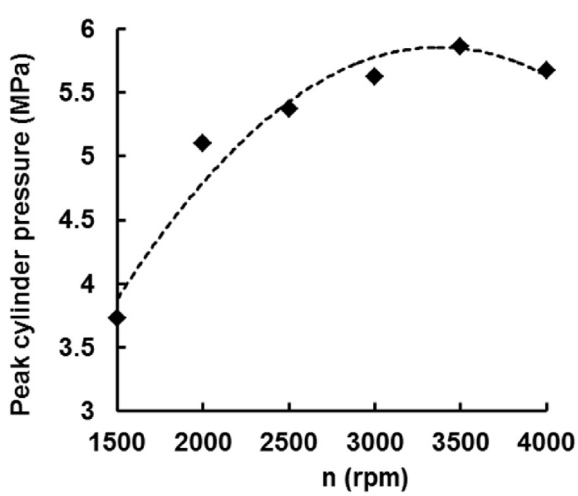

Fig. 4 – The peak cylinder pressure changes with engine speeds at WOT.

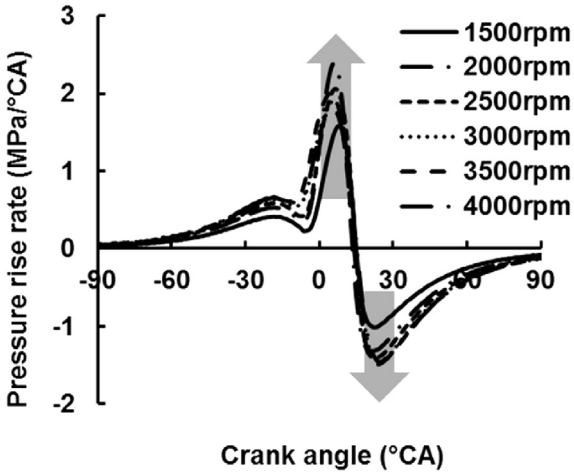

Fig. 5 – The cylinder pressure rise rate changes with crank angle at different engine speeds, WOT.

characterize the combustion process [26,27]. As Fig. 5 shown, the tendency of the pressure rise rate can be divided into two segments according to the different combustion process. First, the combustion pressure rises rapidly from the spark timing to the peak cylinder pressure rise rate, which leads to the increase of the pressure rise rate. The piston downward movement decreases the pressure in the cylinder rapidly. It's the reason that the pressure rise rate appears to be negative as shown in Fig. 5. As Fig. 6 shown, the peak cylinder pressure rise rate obtains the maximum as 0.25 MPa/°CA at 2000 rpm, which means the noise of the turbocharged hydrogen engine is very small when the equivalence ratio is lower than 0.55. The reason can be explained from two aspects. When the engine speed is smaller than 2000 rpm, the exhaust gas energy is inadequate and the turbine doesn't work well, which will increase the back pressure. Moreover, the worse gas exchange process will lead to the increase of the residual gas fraction, which will deteriorate the combustion process and reduce the combustion pressure. The result is that the peak cylinder pressure rise rate is lower than the naturally aspirated engine, which is an obvious difference between the turbocharged and naturally aspirated engine. On the other hand, the peak cylinder pressure rise rate also decreases with increase of the engine speed after 2000 rpm. The reason is that the boost pressure rises and the effect of gas exchange process weakens gradually. At the same time, the time corresponding to the crank angle per degree decreases, which lead to the decrease of pressure rise rate by crank angle per degree.

In addition to the above pressure-related parameters, the combustion parameters are also important for characterizing the combustion process and can provide support for improving the combustion process. With the increase of the engine speeds, the combustion duration increases from 12.5°CA at 1500 rpm to 18.1°CA at 4000 rpm as shown in Fig. 7. The reason is that the time corresponding to the crank angle per degree decreases with the increase of engine speeds, which lead to the combustion duration increases. Moreover, the mass of air flow into the cylinder increases with the increase of engine speeds, especially after the turbocharging. The volumetric coefficient rises, which leads to an increase in air flow mass at high speeds and the combustion duration also increases when the equivalence ratio keeps constant. However, the combustion duration at 1500 rpm and 2000 rpm is basically the same. The reason is that the turbine becomes a source of resistance and blocks the exhaust gas, which increases the residual gas friction and slows down the combustion velocity.

Engine speed also has effect on the NOx emission as Fig. 8 shown. The NOx emission increases first and then descends with the increase of the engine speed and the biggest value is around 1200 ppm at 2500 rpm. The reason can be explained from three aspects. First, there are three factors affecting NOx include: temperature, oxygen concentration and reaction time. From Fig. 2, the combustion pressure increase with the increase of engine speed, which means the combustion temperature also increase. High temperature becomes the dominant factor causing the emission of NOx increases when the engine speed changes from 1500 rpm to 2500 rpm. However, the reaction time becomes a more important factor when the engine speed changes from 2500 rpm to 4000 rpm. The time of

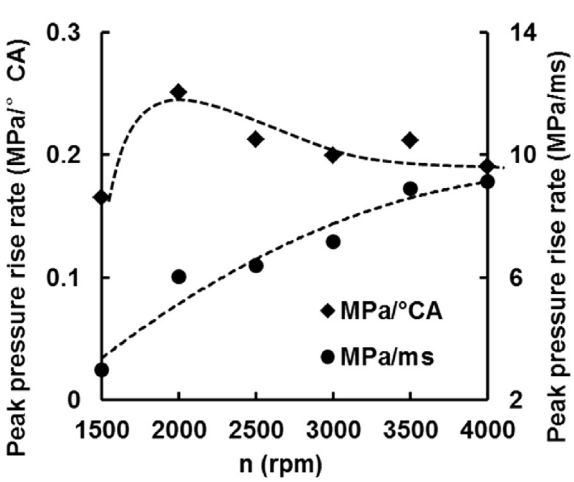

Fig. 6 – The crank angle of the peak pressure sie rate changes with engine speeds at WOT.

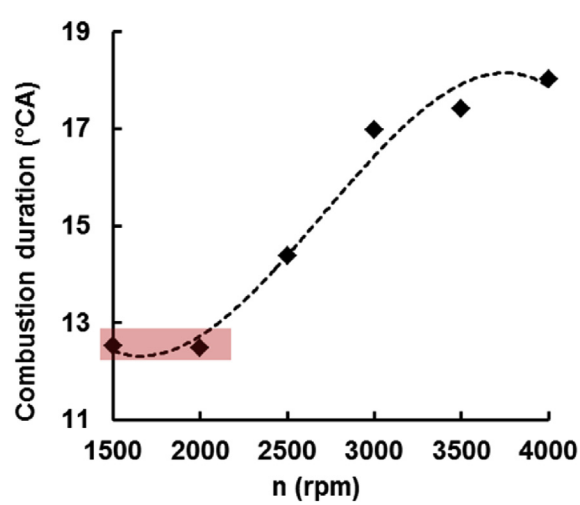

Fig. 7 – The combustion duration changes with engine speeds at WOT.

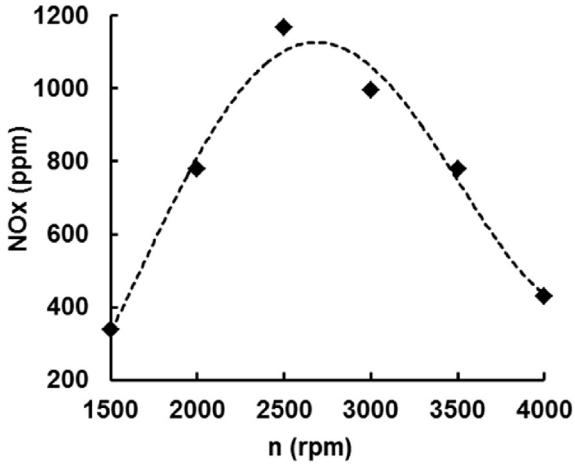

Fig. 8 – The NOx emission changes with engine speeds at WOT.

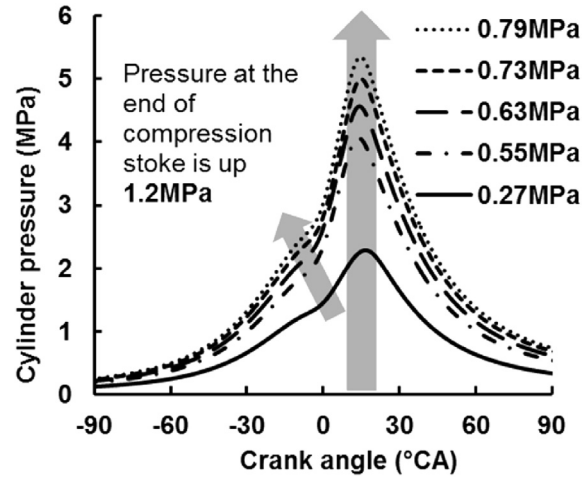

Fig. 10 – The cylinder pressure changes with crank angle at different BMEP, 2500 rpm, $\lambda = 0.55$.

per cycle decreases with the increase of engine speed, which will inhibit the NOx emission. Second, due to the limitation of the injection pressure of the nozzle, the mass flow rate of hydrogen injection is not enough so that the equivalence ratio decreases with the increase of the engine speed, which is a reason of the decrease of NOx as Fig. 9 shown. Third, the boost pressure also increases very small limited by the knock, which also influences the NOx emission. The tendency shown in Fig. 8 can be attributed to these three factors.

Effect of load variation

To explore the effect of the loads variation on the combustion process and NOx emission, the working condition of 2500 rpm, $\lambda = 0.55$, MBT and different loads are chosen. The pressure at the end of compression stoke in the cylinder increases with the increase of loads from 1.2 MPa when the BMEP is 0.27 MPa–2.3 MPa when the BMEP is 0.79 MPa, as shown in Fig. 10. The reason is that volumetric efficiency ascends rapidly with the increase of loads because the throttling of intake air flow decreases. Moreover, the increase of intake air flow mass will lead to the increase of exhaust gas energy, which will boost the pressure of intake air. The result is that the volumetric efficiency further increases and the pressure at the end of compression stoke also increases dramatically. This reason also can be used to explain the results of the peak cylinder pressure rise rate as shown in Fig. 11. The value of the peak cylinder pressure rise rate changes from 0.07 MPa/°CA to 0.22 MPa/°CA.

The tendency of instantaneous heat release changing with loads is similar but the maximum instantaneous heat release increase and the crank angle is constantly close to the TDC as Fig. 12 shown. The instantaneous heat release reflects the combustion process in cylinder and the tendency also can be explained by the increase of the volumetric efficiency as well as intake air mass. However, the combustion duration at different engine speeds occurs around 14°CA except the one when the BMEP is 0.27 MPa, whose value can reach to 16.5°CA as Fig. 13 shown. The reason is that the residual gas friction increases with the decrease of the loads, especially for the turbocharged hydrogen engine. The low loads cause the

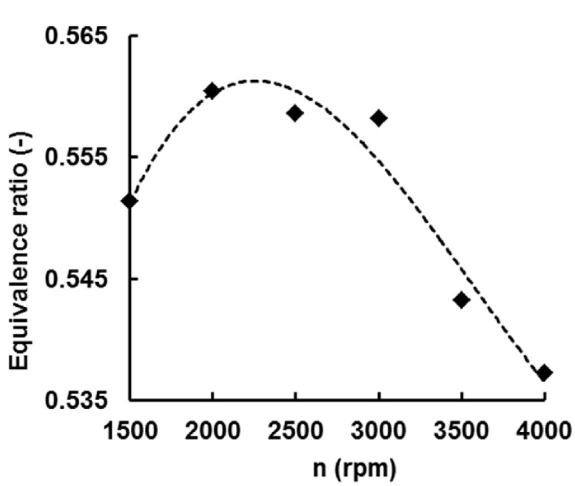

Fig. 9 – The equivalence ratio changes with engine speeds at WOT.

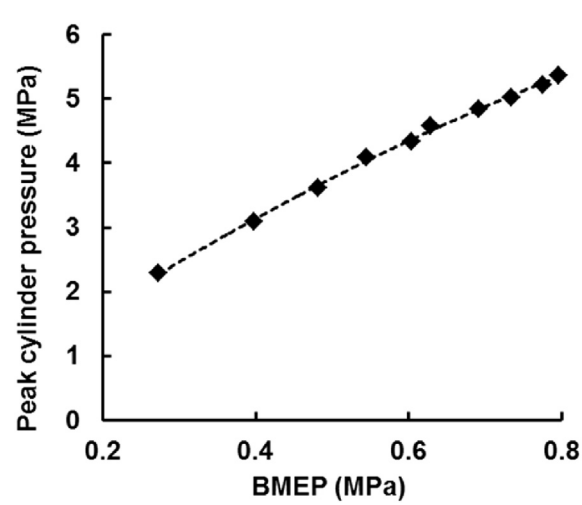

Fig. 11 – The peak cylinder pressure rise rate changes with BMEP at 2500prm, $\lambda = 0.55$.

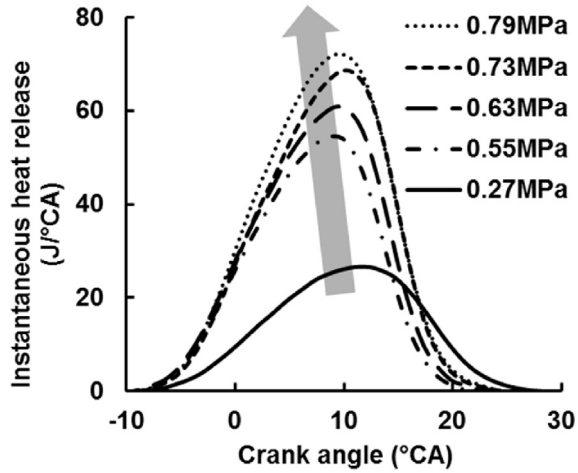

Fig. 12 – The instantaneous heat release changes with crank angle at different BMEP, 2500prm, λ = 0.55.

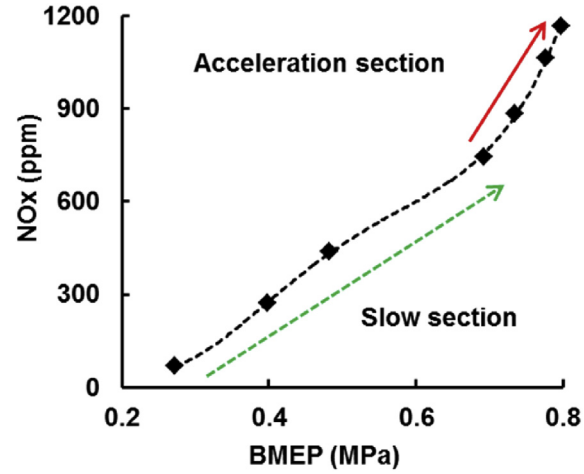

Fig. 14 – The NOx emission changes with BMEP at 2500prm, λ = 0.55.

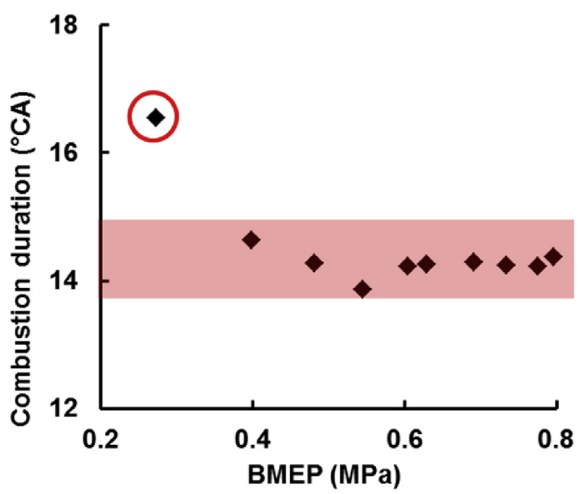

Fig. 13 – The combustion duration changes with BMEP at 2500prm, λ = 0.55.

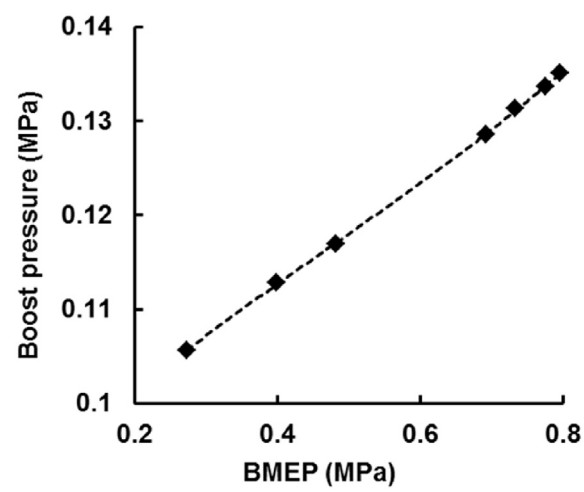

Fig. 15 – The intercooled and boost pressure changes with BMEP at 2500prm, λ = 0.55.

increase of the back pressure because the turbine cannot get the enough energy, which leads to the worse gas exchange process and the higher residual gas friction. The result is that the combustion velocity drops and the combustion duration increases. Thus, it's wise to choose the working conditions according to the power needs to avoid low load areas, particularly for the turbocharged hydrogen engine.

The loads can increase the combustion temperature and the energy absorbed by the turbine increases with the increase of the loads, which will ascend the combustion temperature further. The combustion temperature can promote the formation of NOx emission. As Fig. 14 shown, the general tendency of NOx emission increases and the maximum value is about 1200 ppm. However, the increase rate of NOx emission can be divided into two periods as the loads increase. When the BMEP changes from 0.27 MPa to 0.69 MPa, the increase rate of NOx emission keeps slow. The emission rises rapidly within the range of 0.69 MPa–0.8 MPa. The reason can be explained from two aspects. On the one hand, the increase of the exhaust gas energy leads to the increase of boost pressure as Fig. 15. The boost pressure ascends the pressure at the end of compression stoke, which will increase the combustion temperature. On the other hand, since the NOx formation follows the Zeldovich mechanism, which means that NOx produced increases exponentially with the increase of combustion temperature with the enough energy that temperature provided. The result is that the NOx emission shows a slow trend at the beginning and accelerates in the next period.

Effect of equivalence ratio

Equivalence ratio is an important parameter to the hydrogen engine because it closely relate to the combustion velocity. To explore the effect of equivalence ratio on the combustion characteristic and NOx emission of turbocharged hydrogen engine, the working conditions at 2500 rpm, MBT, throttle angle of 17 deg and λ = 0.4–1.1 interval of 0.1 are chosen. The combustion pressure increases as the equivalence ratio

increases, as shown in Fig. 16. The reason is that the pressure behind intercooler increases with the increase of equivalence ratio as Fig. 17 shown, which will improve the gas exchange process and increase the volumetric efficiency. The higher volumetric efficiency leads to the more intake flow mass. Therefore, the pressure at the end of compression stoke will increase and the combustion pressure also go up drastically.

The peak cylinder pressure rise rate increases as the equivalence ratio increase as shown in Fig. 18 because the bigger equivalence ratio means the faster combustion velocity, which will lead to the increase of pressure rate per unit time. As Fig. 19 shown, the crank angle of the peak cylinder pressure rise rate deviates from the TDC as the increase of equivalence ratio which means the lower isometric of Otto cycle. The lower isometric of Otto cycle means the lower BTE. It also proves that the lean equivalence ratio can increase the BTE of hydrogen engine. The instantaneous heat release increases sharply as the increase of equivalence ratio and its maximum value is more than 7 times more than that of 0.4 when the equivalence ratio is equal to1 as shown in Fig. 20.

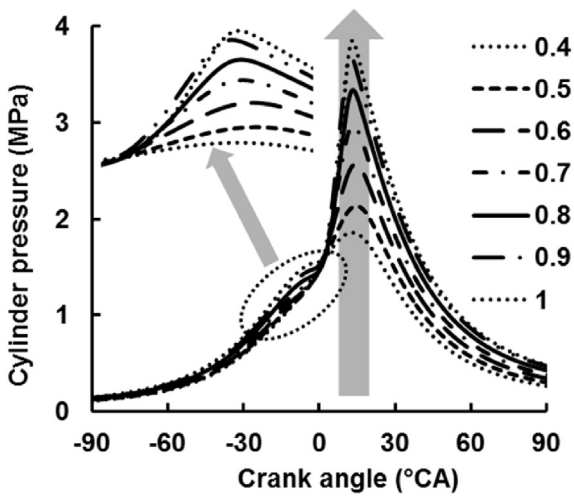

Fig. 16 – The cylinder pressure changes with crank angle at different equivalence ratios, 2500 rpm, WOT.

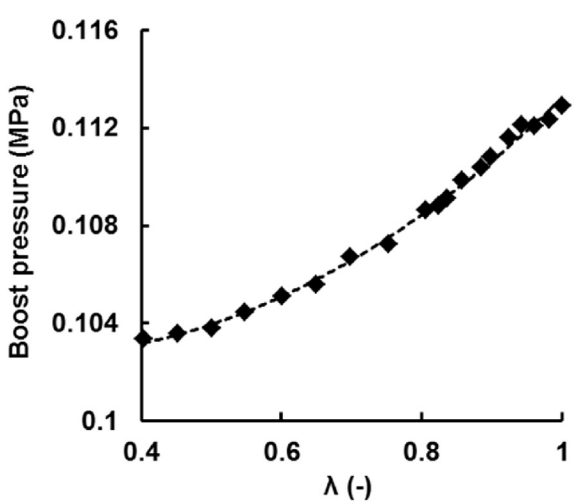

Fig. 17 – The intercooled and boost pressure changes with λ at 2500prm, WOT.

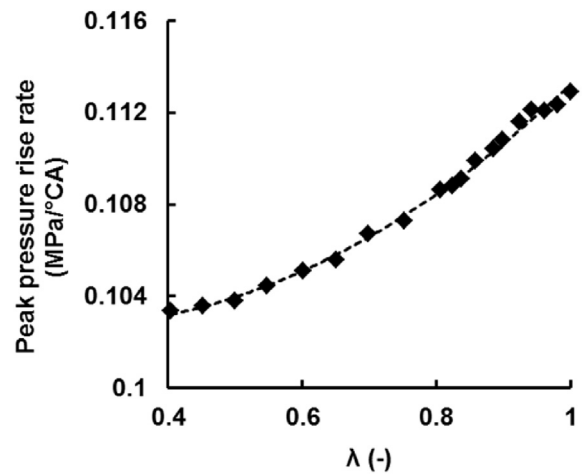

Fig. 18 – The peak cylinder pressure rise rate changes with λ at 2500prm, WOT.

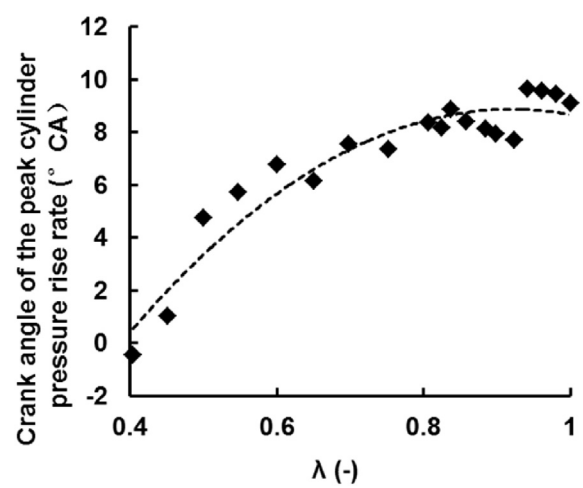

Fig. 19 – The crank angle of the peak cylinder pressure rise rate changes with λ at 2500prm, WOT.

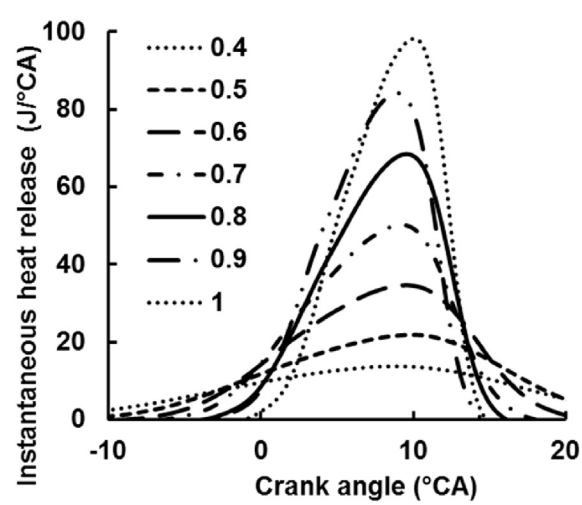

Fig. 20 – The instantaneous heat release changes with crank angle at different equivalence ratios, 2500prm, WOT.

The reason is that the bigger equivalence ratio leads to the larger air intake mass, which will increase the instantaneous heat release. Moreover, the combustion duration decreases with the increase of equivalence ratio, which will further increase instantaneous heat release as Fig. 21 shown. The equivalence ratio can decrease the combustion duration, which means that the leaner equivalence ratio is the lower combustion velocity when the working conditions keep constant. The result is that the benefit of BTE from the lean equivalence ratio is limited by the combustion velocity. Thus, using the lean equivalence ratio to increase BTE should consider the effect of combustion velocity on the combustion duration. Comprehensive consideration of the influence law of two factors can obtain the highest BTE for turbocharged hydrogen engine.

Equivalence ratio has effect on the NOx emission because it influences the combustion velocity. The faster combustion velocity means the higher temperature and the NOx emission will increase. Moreover, the increase of exhaust gas energy increases the intake air pressure, which also increases the NOx emission. The tendency of NOx emission with the equivalence ratio can be divided into 4 sections: slow up section at $\lambda < 0.6$, fast up section at $0.6 \leqq \lambda \leqq 0.88$, fast down section at $0.88 < \lambda \leqq 1$ and slow down section at $\lambda > 1$ as Fig. 22 shown. The reason also can be explained by the Zeldovich mechanism. As Fig. 23 shown, the maximum combustion temperature is lower than 1600 K when $\lambda < 0.6$, which cannot provide sufficient activation energy. The NOx emission is lower than 200 ppm in slow up section because the temperature is the dominant factor. When the equivalence ratio changes from 0.6 to 0.88, the combustion temperature increase from 1600 K to 1800 K and the oxygen mass concentration is bigger than 4%, which is the comfortable condition for producing the NOx emission. The NOx emission increases from 200 ppm to around 7000 ppm in the fast up section. When the equivalence ratio changes from 0.88 to 1, although the combustion temperature is higher than 1800 K, the oxygen mass concentration reduces greatly, which will reduce the probability of chemical reaction between nitrogen and oxygen in the cylinder. The result is the NOx emission in the fast down section decreases from around 7000 ppm to 1000 ppm. When $\lambda > 1$, although the oxygen mass concentration is close to 0, the NOx emission still exists in the exhaust gas due to the uneven composition and the uneven temperature distribution, which decreases with the increase of the equivalence ratio.

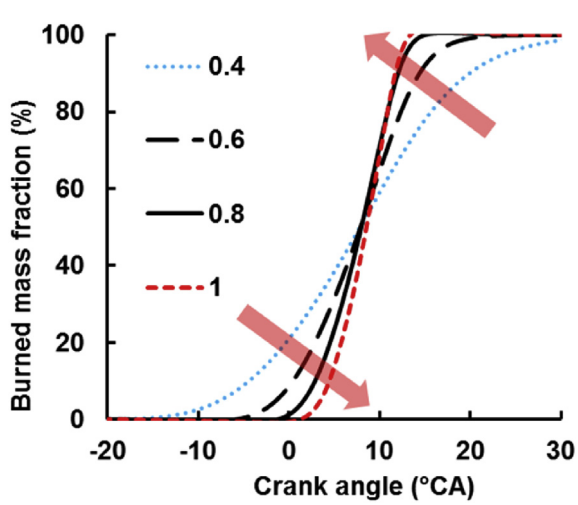

Fig. 21 – The burned mass fraction changes with crank angle at different equivalence ratios at 2500prm, WOT.

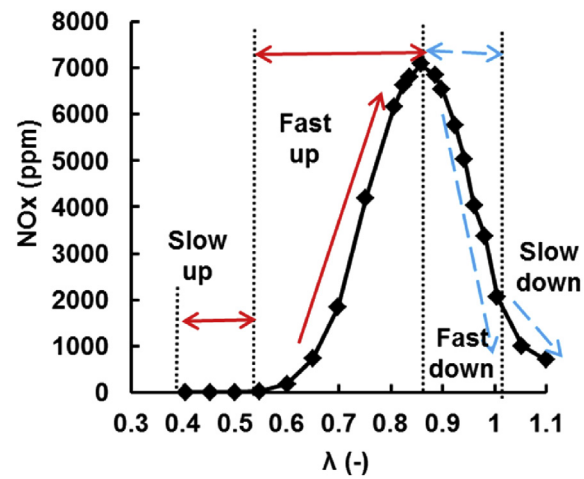

Fig. 22 – The NOx emission changes with λ at 2500 rpm, WOT.

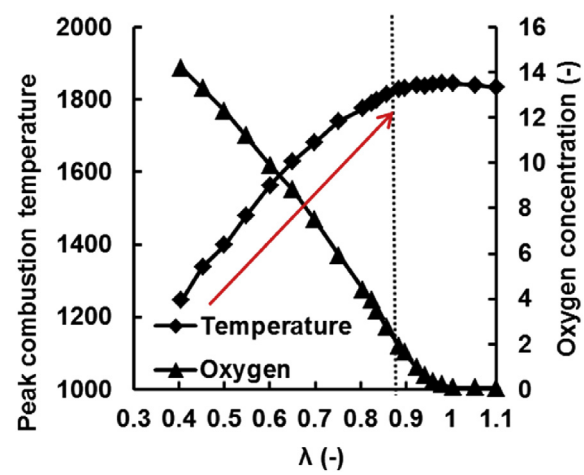

Fig. 23 – The maximum combustion temperature and oxygen mass concentration changes with λ at 2500prm, WOT.

Effect of spark timing

Spark timing influences the combustion characteristic and NOx emission duo to its effect on the combustion velocity. To avoid the possibility of backfire at WOT, the test was conducted at 2500 rpm, $\lambda = 0.55$, 75%, 50% and 25% of loads. As Figs. 24 and 25 shown, retarding the timing decreases the peak cylinder pressure and corresponding crank angle continuously deviates from the TDC because more fuel burns after TDC. When the ST changes from 30°CA before TDC to TDC, the peak cylinder pressure decreases from 3.3 MPa to 1.7 MPa and the crank angle changes from 2.5°CA after TDC to 30°CA after

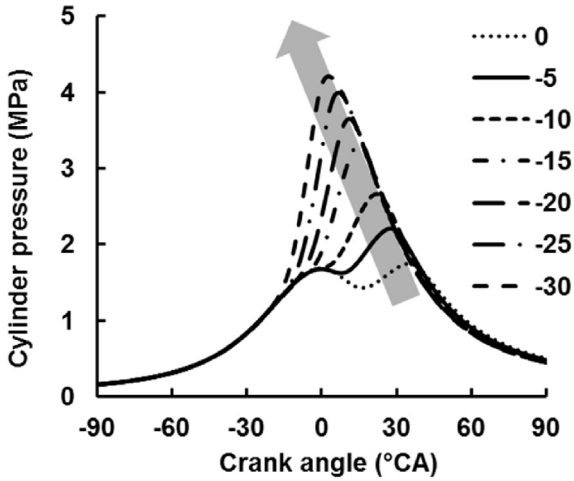

Fig. 24 – The cylinder pressure changes with crank angle at different ST, 2500 rpm, WOT, λ = 0.55.

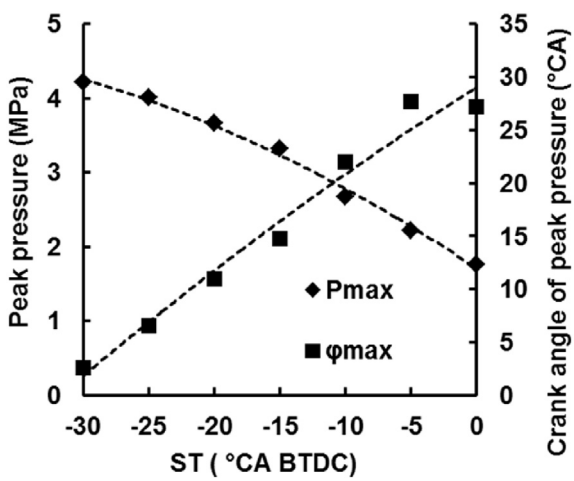

Fig. 25 – The peak cylinder pressure and corresponding crank angle changes with ST, 2500 rpm, WOT, λ = 0.55.

TDC. The result means that the earlier ST, the bigger peak cylinder pressure, which means that the bigger power.

However, the peak cylinder pressure is just a parameter to characteristic the combustion process in the cylinder. Moreover, advancing timing increases the negative works in the compression stoke. Thus, more parameters are needed to explore the effect of ST on the combustion process. As Fig. 26 shown, the peak cylinder pressure rise rate decrease as the ST is closely to the TDC, which means that the combustion noise also decreases. Moreover, advancing timing will lead to the increase of the peak cylinder pressure rise rate, which will decrease the BTE as Fig. 27 shown. Retarding the timing will increase the combustion duration as Fig. 28 shown. The reason is that the temperature and pressure of mixture increase when the position pushes the mixture, which can increase turbulence intensity and combustion velocity. The result is the combustion duration shortens with the early ST. Burned mass fraction is also an important parameter to evaluate the combustion process. As Fig. 29 shown, the curve of the burned mass fraction is constantly panning to the right

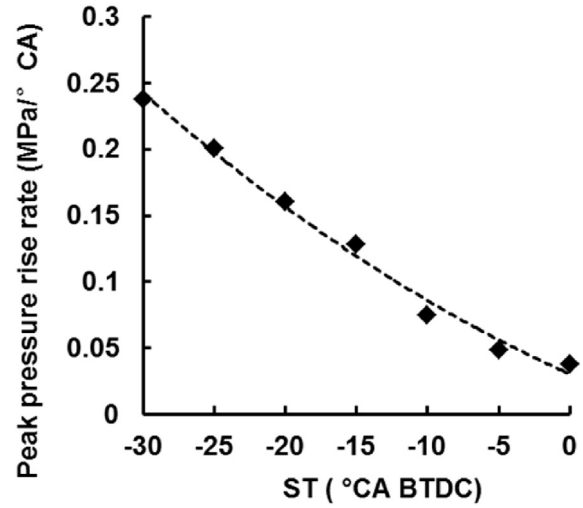

Fig. 26 – The peak cylinder pressure rise rate changes with ST, 2500 rpm, WOT, λ = 0.55.

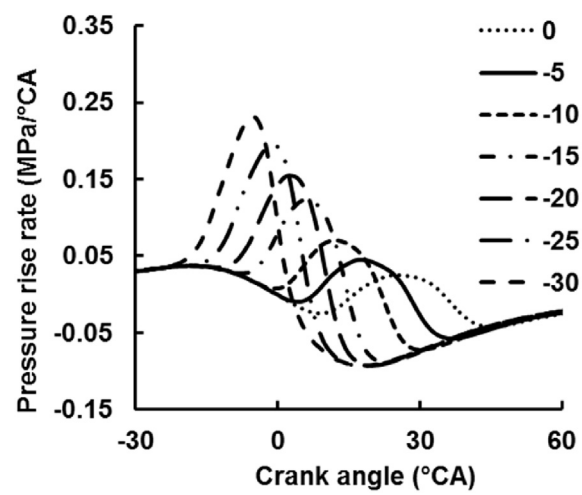

Fig. 27 – The cylinder pressure rise rate changes with crank angle at different ST, 2500 rpm, WOT, λ = 0.55.

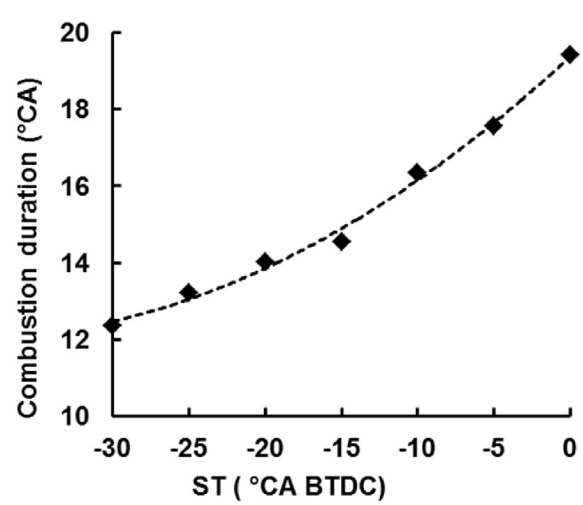

Fig. 28 – The combustion duration changes with ST, 2500 rpm, WOT, λ = 0.55.

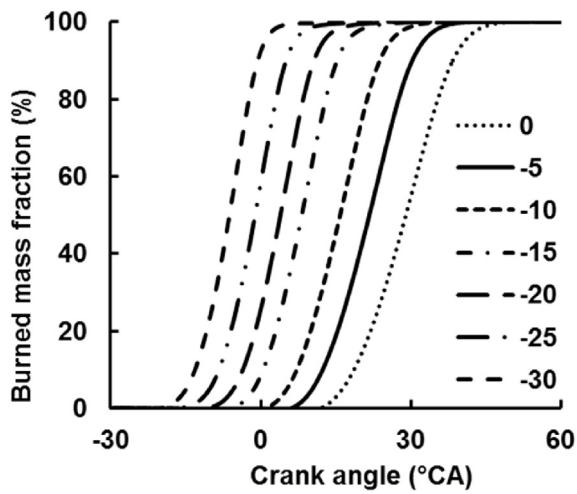

Fig. 29 – The burned mass fraction changes with crank angle at different ST, 2500prm, WOT, λ = 0.55.

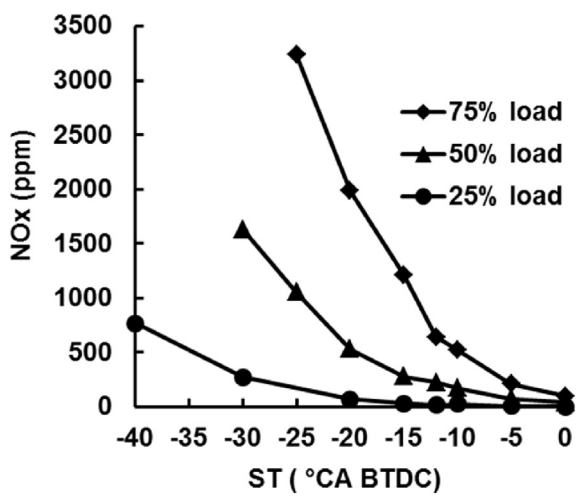

Fig. 30 – The NOx emission changes with ST at different loads, 2500prm, WOT, λ = 0.55.

(close to TDC and then away TDC) as the ST close to the TDC. According to the principle of Otto cycle, the more heat is released around the TDC, the higher BTE can be got. Thus, the optimized ST should be 15 °CA before TDC for the BTE. Analyzing the effect of ST to the combustion process, it can be found that the optimized ST can be chosen according to the power or the BTE need at different work conditions.

The ST also has effect on the NOx emission as Fig. 30 shown. The NOx emission decreases with the ST close to the TDC at different loads and shows an exponential trend. The reason is that advancing the ST will increase the peak cylinder pressure, which will increase the temperature.

Conclusion

Effect of different parameters including engine speeds, loads, equivalence ratios and STs on the combustion characteristic and NOx emission of the turbocharged hydrogen engine has been investigated using the test data in this paper. The conclusions can be found as following:

(1) The turbocharger influences the combustion duration tendency with engine speeds. When the turbine cannot absorb enough exhaust gas energy, the residual gas fraction increases and the combustion velocity slows down, which leads to the equal of the combustion duration at 1500 rpm and 2000 rpm. The result at 2500 rpm, λ = 0.55 and different loads also can prove the conclusion. This is an obvious difference of combustion characteristic between the turbocharged and naturally aspirated engine.
(2) The peak cylinder pressure and the rise rate both increase with the increase of the engine speeds, load and equivalence ratio, but decrease with ST close to the TDC. The peak cylinder pressure rise rate at all working conditions are lower than 0.25 MPa/°CA, which means the combustion noise of turbocharged hydrogen engines can reach to the standard of gasoline engines.
(3) The NOx emission shows the tendency of increase first and then decrease with the increase of engine speeds and the range is 300ppm-1200 ppm. The NOx emission presents two sections of slow up and fast up with the increase of loads because the boost pressure increase the combustion temperature. Moreover, the effect of equivalence ratio on the NOx emission can be divided into 4 parts according to the temperature and oxygen mass concentration and the maximum value can reach to 7000 ppm, which is much higher than the gasoline engines. Finally, the NOx emission decreases rapidly as the ST close to the TDC because the peak cylinder pressure decreases.

Acknowledgement

The financial support of the National Natural Science Foundation of China (NSFC 51276019) is greatly acknowledged.

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